

Yazhuo Liu

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Education

Georgia Institute of Technology , Atlanta, USA	Jan. 2022 – Now
<i>Ph.D. in Mechanical Engineering</i> (mechanics of materials)	
Southern University of Science and Technology , Shenzhen, China	Sep. 2017 – Jun. 2021
<i>B.Sc. in Theoretical and Applied Mechanics</i> (solid mechanics)	

Experience

Georgia Institute of Technology , Atlanta, Georgia, USA	Aug. 2025 – May. 2026
<i>Instructor. Teaching COE3001 - Deformable Bodies</i>	
Southern University of Science and Technology , Shenzhen, China	Jul. 2021 – Dec. 2021
<i>Research Assistant. Supervisor: Prof. Wei Hong</i>	
Cornell University , Ithaca, New York, USA (Online due to COVID-19)	Jun. 2020 – Sep. 2020
<i>Summer Research Intern. Advisor: Prof. Chung-Yuen Hui</i>	

Publications

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- Y. Liu, Y. Zhang, K. Ding, Y. Yang, A. Barrios, X. Maeder, O. Pierron, X. Liu, & T. Zhu. Phase-field modeling of elastically driven abnormal grain growth. *Journal of the Mechanics and Physics of Solids* (in progress).
- Y. Yang, A. Barrios, Q. Li, Y. Liu, M. Jain, V. Meyers, T. Zhu, B. L. Boyce, & O. Pierron. Elucidating the origins of grain growth in the high cycle fatigue degradation of Au thin films. *Acta Materialia*, 313, 122301 (2026).
- Y. Yang, Y. Liu, T. Zhu, O. Pierron, & J. Kacher. Decoupling the effects of temperature and stress on grain growth in nanocrystalline Ni films. *Acta Materialia*, 313, 122305 (2026).
- Y. Liu, K. Ding, A. J. Birnbaum, A. Rawlings, A. Sun, W. Chen, D. L. McDowell, J. G. Michopoulos, & T. Zhu. Surface plasticity in laser scanning of metals. *Acta Materialia*, 302, 121667 (2026).
- Y. Liu, X. Feng, & W. Hong. Non-affine dissipation in polymer fracture. *Extreme Mechanics Letters*, 59, 101955 (2023).
- Y. Liu, C.-Y. Hui, & W. Hong. A clean cut. *Extreme Mechanics Letters*, 46, 101343 (2021).

Patents

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- CN Patent No. CN202020206173.X: “Logistics Unmanned Aerial Vehicle”, October 27, 2020. (Shared)

Presentations

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- TMS 2024 Annual Meeting & Exhibition** - oral presentation. “Crystal plasticity modeling of thermo-elastic-plastic deformation during laser-based additive manufacturing”, Orlando, Florida, USA
- 2025 SES Annual Technical Meeting** - oral presentation. “Abnormal grain growth under cyclic loading”, Atlanta, Georgia, USA
- 2025 Physical Metallurgy Gordon Research Conference** - poster. “Driving force of abnormal grain growth”, Easton, Massachusetts, USA
- TMS 2026 Annual Meeting & Exhibition** - oral presentation. “Phase field modeling of abnormal grain growth”, San Diego, California, USA

Awards & Honors

- *SUSTech Summa Cum Laude Graduates*, 2021.
- *COMAC 3D Printing Structure Optimization Competition*, Third Prize, 2018.

Projects

Dislocation-defect interactions Aug. 2025 - now

- The objective is to study the interaction between dislocations and other objects in BCC crystals using molecular dynamics (MD), molecular statics (MS) and nudged elastic band (NEB) methods.
- Project still in progress

Stress induced phase separation of Li dendrite in LLZTO solid-state electrolyte Apr. 2025 - now

- Identified an early-stage, mesh-pattern morphology of Li dendrite growth in LLZTO under low-current short-circuit conditions through experimental observations.
- Developed a mechanistic theoretical modeling framework to describe the self-organization of Li, and to understand the evolution of dendritic network structures.

Laser-scanning-induced plastic deformation in metal 3D printing Apr. 2023 - Aug. 2025

- Investigated the influence of laser scanning on stress and plastic strain evolution in Ni single crystals, visualizing plastic deformation through surface slip traces and dislocation patterns.
- Developed a temperature-dependent crystal plasticity finite element (CPFE) model to simulate thermal effects and plastic deformation during laser scanning along various directions.
- Validated the CPFE model with experimental data, providing insights into controlling thermo-mechanical responses in laser-based additive manufacturing.

Abnormal grain growth in ultra-fine grained metals Sep. 2022 - Aug. 2025

- Developed a micromechanics-based analytical solution to predict grain boundary motion in thin-film metals using local stress-strain and crystallographic data.
- Utilized continuum phase field modeling to simulate multi-grain systems, capturing spatial and temporal evolution of grain structures under various loading conditions.
- Validated the phase field model through comparisons with analytical predictions and experimental data, demonstrating its accuracy in predicting grain growth behavior.

Energy dissipation mechanism in polymer fracture Feb. 2021 – Dec. 2021

- Revealed a significant energy dissipation mechanism in polymeric fracture, attributed to non-affine deformation near the crack tip, beyond traditional viscoelastic hysteresis.
- Demonstrated that reducing inter-chain friction through methods like lateral stretching or solvent dilution lowers fracture energy to intrinsic thresholds, while cyclic training maintains high fracture toughness.
- Identified non-affine dissipation as a key factor for developing soft materials with superior elasticity and fracture resistance, offering new design principles for advanced materials.

Effect of Friction in Cutting Soft Solids Jun. 2020 – Jun. 2021

- Investigated the role of friction and slicing motion in cutting soft solids, revealing how friction influences stress distribution and fracture initiation during the process.
- Developed a numerical model to analyze frictional contact and stress diversion, demonstrating how slicing minimizes vertical resistance and facilitates clean cutting.
- Studied the energetics of cutting by computing the energy landscape, showing that friction reduction lowers the energy barrier for crack opening and enhances cutting efficiency.